



USER MANUAL & SYSTEM GUIDE

Donna Barbar Polytechnic College Management System

A complete, plain-language guide to every module, entity, role and workflow in the DBPCMS platform — for administrators, staff and academic teams.

Donna Barbar Polytechnic College · Oromia, Ethiopia

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I

Orientation

Understanding what DBPCMS is, why it exists, and the ideas that run through every part of it.

CHAPTER 1

Welcome & How to Use This Manual

This handbook is the complete, plain-language companion to the Donna Barbar Polytechnic College Management System. It explains not only how to use every screen, but why each part exists and how the pieces fit together.

The Donna Barbar Polytechnic College Management System — DBPCMS for short — is the digital backbone of the college. It replaces scattered spreadsheets, paper files and disconnected registers with one secure, organised place where staff records, students, courses, grades and official documents all live together. Before you touch a single button, it helps to understand what this manual will give you: a guided tour, written for a real person rather than a computer engineer, that takes you from your very first login all the way to producing an official, verifiable transcript.

1.1 The purpose of DBPCMS

A polytechnic college is a busy organisation. In any given week the registrar is enrolling students, instructors are recording quiz marks, the human-resources office is updating an employee's contract, a department head is approving results, and the dean is asking how many students passed a given course. Each of those activities generates information, and every piece of information needs to be accurate, findable and trustworthy. DBPCMS exists to make that possible. It is a **single source of truth**: when a fact is recorded once — a student's name, a course's credit hours, an employee's date of employment — it is stored in one place and reused everywhere, so nobody has to re-type it and no two copies can disagree.

1.2 Who this manual is for

Everyone who logs into DBPCMS will find their part of the story here. The manual is deliberately written so that a brand-new employee can read it from cover to cover and understand the whole system, while an experienced administrator can jump straight to the chapter they need. Wherever a task belongs to a specific role — say, publishing grades, which only the registrar can do — the manual says so clearly.

System Administrator	Read every chapter; you are responsible for settings, users, backups and the overall health of the platform.
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HR Officer	Focus on Chapters 8 (Employees), 13 (Reports) and 14 (Settings).
Registrar	Chapters 7, 9, 11 and 12 are your daily companions.
Department Head	Chapters 7, 10 and 11 — especially grade approvals.
Instructor	Chapter 11 (grade entry) is the one you will use most.
Dean & Leadership	Chapters 6 and 13 for dashboards and analytics.

1.3 Conventions used throughout

To keep things easy to scan, the manual uses a small set of visual signals. When you see a coloured box, it is drawing your attention to something worth pausing on:

Note

Background information or context that helps a task make sense. Nothing will break if you skip it, but you will understand the system more deeply if you read it.

Tip

A shortcut, a good habit, or a smarter way of doing something that experienced users rely on.

Important

Something you must not overlook — usually because it affects data that other people depend on, or because it cannot easily be undone.

Configurable

A value that is **not** fixed in the software but is set by an administrator on a settings screen. One of the guiding rules of DBPCMS is that business rules live in the interface, not buried in code.

Technical names — the exact label of a field, a button, or an identifier — are shown in a `mono-space font` so you can recognise them instantly.

CHAPTER 2

What DBPCMS Is & What It Does

At heart, DBPCMS is two great responsibilities — managing people who work at the college and managing the academic life of its students — joined by a shared spine of security, structure and record-keeping.

2.1 A single, connected system

Think of the college as a living organism. The staff are its hands, the students its purpose, the departments its skeleton, and the flow of grades and records its circulation. DBPCMS models all of this faithfully. A department is not just a label on a form — it is a real container that owns programs, employs staff, offers courses, and enrolls students. Because these relationships are built into the very foundation of the system, information stays consistent: change a department's name once and it updates everywhere that department appears, from an employee's profile to a student's transcript.

This connectedness is the single biggest advantage of DBPCMS over the paper-and-spreadsheet approach it replaces. In a spreadsheet world, the same student might appear in five different files with three different spellings of their name. Here, that student is *one record*, referenced everywhere, always in agreement with itself.

2.2 The two headline modules

Employee Management

The Employee Management module is the college's complete human-resources record. Every member of staff — teaching or non-teaching — has a rich profile that captures their personal details, their position and department, their employment history, their academic qualifications, their emergency contacts, and their scanned official documents. From this data the system produces a printable, officially-styled staff profile carrying a QR code that anyone can scan to confirm the document is genuine. HR officers can generate seven kinds of reports — from a full staff directory to a list of employees approaching retirement — and export any of them to PDF, Excel or CSV.

Student Grading

The Grading module is where the academic year comes to life. Instructors record marks for each assessment — a quiz, an assignment, a mid exam, a final exam — and the system automatically combines them, applies the college's grading scale, and produces a letter grade, a grade point and a pass/fail outcome. Crucially, grades do not simply appear on a transcript the moment they are typed. They travel through a deliberate, auditable workflow: an instructor enters and submits them, a department head reviews and approves them, and only the registrar can finally publish them. Once published, a grade is frozen — a permanent, tamper-proof record.

2.3 The principles that shaped the design

Configurable, not coded	Grading scales, GPA rules, retirement age, ID formats and more are all set in the interface. The rules of the college can change without any programming.
Permission-based security	The system checks what you are <i>allowed to do</i> , not merely what your job title is. This makes access precise and easy to adjust.
Nothing is truly deleted	Records are archived (soft-deleted) rather than destroyed, so history is never lost and mistakes can be recovered.
Everything is remembered	Every meaningful change is written to an audit log — who did what, when, and what the value was before and after.
Built for two languages	The interface ships in English and is prepared for Afaan Oromo, reflecting the college's community.

Configurable

Throughout this manual you will meet this box. Each time, it marks a rule that an administrator can change on a settings screen — a promise that the software bends to the college, not the other way around.

CHAPTER 3

Core Concepts & Vocabulary

A handful of ideas appear again and again across every module. Understanding them once, here, makes the rest of the manual effortless.

3.1 Entities: the "things" the system keeps track of

An **entity** is simply a type of thing that the system stores information about. A department is an entity; so is an employee, a student, a course and a grade. Each individual example of an entity — the Information Technology department, the student Kebede Alemu, the course "Introduction to Programming" — is called a **record**. Much of your work in DBPCMS consists of creating, viewing, editing and organising records.

Records are connected to one another through **relationships**. A program belongs to a department. A section belongs to a course and is taught in a semester. A student is enrolled in many sections. These relationships are what turn a pile of separate facts into a coherent picture of the college. The table below lists the main entities you will encounter and what each one represents.

Entity	What it represents
Department	An academic or administrative unit of the college that owns programs, employs staff and offers courses.
Program	A qualification a student studies toward — for example a Diploma or Degree — belonging to one department and lasting a set number of years.
Academic Year	A named year of study (e.g. 2026) with a start and end date; exactly one is marked "current".
Semester	A teaching period within an academic year, with its own dates and status (planned, active or closed).
Course	A subject with a code, title and credit hours, optionally tied to a program.
Section	A specific offering of a course in a specific semester — the class that instructors teach and students enrol in.
Employee	A member of staff, with a full personal and professional profile.

Entity	What it represents
Student	A learner enrolled at the college, belonging to a department and program.
Enrollment	The link that places one student in one section — the thing a grade is ultimately attached to.
Grade Component	One assessment type (Quiz, Assignment, Mid Exam, Final Exam) with a weight and a maximum score.
Grading Scale	The set of bands that turn a percentage into a letter grade and grade point.
Document	A scanned or uploaded file (degree, contract, ID...) attached to an employee.
User	A login account, granted one or more roles, that a person uses to access the system.

3.2 Roles versus permissions

This is the single most important security idea in DBPCMS, and it is worth reading slowly. A **permission** is the right to perform one specific action — for example "enter grades", "publish grades", or "create an employee". A **role** is simply a named bundle of permissions given to people who do a similar job — for example the *Registrar* role bundles together the permissions needed to manage students, sections and grade publishing.

The clever part is that the software always checks *permissions*, never role names. When the system decides whether to show you the "Publish" button, it does not ask "is this person a registrar?" — it asks "does this person have the `grade:publish` permission?" This means the college can redesign what a role can do, or invent an entirely new role, without any changes to the software itself. It is the professional way to build access control, and it keeps the system flexible for years to come.

Note

A single user can hold more than one role. An instructor who also heads a department, for instance, would carry both the *Instructor* and *Department Head* roles, and would therefore have the combined permissions of both.

3.3 The safety net: soft deletes, versions and the audit log

Three quiet features protect your data at all times. First, **soft deletes**: when you "delete" most records, DBPCMS actually marks them as archived and hides them, rather than erasing them. The history remains intact and recovery is possible. Second, **optimistic locking**: every record carries a hidden version number, so if two people edit the same record at the same time the system notices and prevents one from silently overwriting the other. Third, the **audit log**: every significant create, update or delete is recorded with the responsible user, the time, and the values before and after the change. Together these mean the system is honest about its own history — nothing important is ever quietly lost or altered.



II

Access & Security

How people get into the system, and how their access is controlled, role by role.

CHAPTER 4

Signing In & Account Security

Security begins at the login screen. This chapter explains how to get into the system safely, what happens on your very first login, and how the platform protects accounts from misuse.

4.1 Your first login

When an administrator creates your account, they set a temporary password and hand it to you securely. The first time you sign in, DBPCMS will insist that you change it. This is not an inconvenience — it is a deliberate safeguard, ensuring that from the very first moment only you know your password. Enter your email address and the temporary password on the login screen, and the system will immediately guide you to the **Change Password** page. Choose a strong new password, confirm it, and you are in.

Tip

Every password field in DBPCMS has a small eye icon. Click it to reveal what you have typed — invaluable for confirming a long password before you commit to it.

4.2 How your session works

Once you are signed in, DBPCMS keeps you logged in using a pair of security tokens working quietly in the background. A short-lived **access token** proves who you are for a few minutes at a time; a longer-lived **refresh token**, stored safely and invisibly, renews it so you are not asked to log in every few minutes. You never see or manage these directly — but it is useful to know why, after a long period of inactivity, you may occasionally be asked to sign in again. It is the system keeping your account safe.

Important

Your permissions are carried inside your session token. If an administrator changes your roles or permissions while you are logged in, you must **log out and log back in** for the change to take effect — otherwise the sidebar and buttons will still reflect your old access.

4.3 Account logout — protection against guessing

To defend against someone trying to guess a password, DBPCMS counts failed login attempts. After several wrong passwords in a short window, the account is temporarily locked and further attempts are refused for a cooling-off period, even if the correct password is then entered. This frustrates attackers while a genuine user simply waits a few minutes or asks an administrator for help.

4.4 Resetting a forgotten password

DBPCMS does not send password-reset emails. Instead, password resets are handled by a person you trust: a System Administrator. If you forget your password, contact your administrator, who will set a new temporary password for your account. Just like your very first login, you will then be required to change it immediately, so the administrator never retains knowledge of your real password.

Note

This human-in-the-loop approach is a deliberate choice for Version 1. It keeps the system simple and avoids depending on email delivery, which can be unreliable — while still ensuring that only the account's owner ever knows the working password.

CHAPTER 5

Roles & the Administration Team

DBPCMS defines eight roles, each shaped around a real job at the college. This chapter introduces every one of them, explains exactly what it can do, and provides a reference you can return to whenever a question of "who is allowed to..." arises.

The college runs on people, and so does DBPCMS. Rather than giving everyone the same keys, the system hands each person exactly the access their job requires — no more, no less. This is the principle of *least privilege*, and it protects both the data and the people who work with it. An instructor cannot accidentally publish grades; a registrar cannot accidentally rewrite an employee's contract; and the powerful ability to change the whole system's settings rests only with those explicitly trusted to hold it.

The eight roles below are seeded into the system automatically. An administrator can create additional custom roles later, mixing and matching permissions as the college's structure evolves — but these eight cover the essential work of running the institution.

System Administrator

Full control of the platform

The System Administrator is the guardian of the entire system. This role holds every permission that exists, which means it can do anything any other role can do — and several things no one else can, such as managing user accounts, editing system settings, reading the audit log, and unlocking or reconfiguring the grading engine. Because this role is so powerful, it should be given to as few trusted people as possible, and those people should be especially careful. The administrator is also the person who creates every other account, assigns roles, resets forgotten passwords, and takes responsibility for backups and the ongoing health of the platform.

Key responsibilities

- Create and manage all user accounts and their roles
- Configure system-wide settings, ID prefixes and business rules
- Read the audit log to investigate any change
- Perform and verify backups; restore data if ever needed
- Step in for any other role when required

Representative permissions

Create user accounts Edit user accounts Reset passwords

Manage roles & permissions Manage system settings Read the audit log

Unlock published grades Configure grading Manage departments Manage programs

Manage academic years Manage semesters Manage courses

Manage sections & enrolment

HR Officer

Keeper of staff records

The HR Officer owns the human side of the college. They create and maintain the full profile of every employee — personal details, position, department, employment history, qualifications, emergency contacts and scanned documents. When leadership needs to know the shape of the workforce, the HR Officer produces the reports: the staff directory, contract-expiry lists, retirement lists and more. They can print the official, QR-verified employee profile that staff use for formal purposes.

Key responsibilities

- Create and update employee records and all their sub-records
- Upload and manage each employee's official documents
- Generate and export the seven HR reports
- Produce printable, verifiable staff profiles

Representative permissions

View employees

Create employees

Edit employees

Print employee profiles

Upload documents

View/download documents

View departments

View & export reports

Dean

Institutional oversight

The Dean sees the college from above. Rather than doing day-to-day data entry, the Dean reads: browsing employees, departments, programs and students, and — most importantly — studying the analytics that reveal how the institution is performing. Pass rates, top students, department GPA comparisons and failure analysis are all within the Dean's view, making this the role for strategy and leadership.

Key responsibilities

- Review the institution's overall performance through analytics
- Browse departments, programs, staff and students for oversight
- Export reports to support leadership decisions

Representative permissions

[View employees](#)[View departments](#)[View programs](#)[View students](#)[View & export reports](#)[View analytics](#)

Registrar

Custodian of the academic record

The Registrar is the heart of the academic administration. They manage the student body, set up semesters, courses and sections, and hold two powers that no instructor or department head possesses: the authority to *publish* grades — turning approved results into permanent, official records — and to *unlock* a published result when a genuine correction is required. The Registrar also generates official transcripts. If the grading workflow is a relay race, the Registrar runs the final, decisive leg.

Key responsibilities

- Enrol and manage students throughout their studies
- Create semesters, courses and sections
- Publish approved grades into the permanent record
- Unlock published grades for authorised corrections
- Generate and issue official transcripts

Representative permissions

View students

Manage students

Manage semesters

Manage courses

Manage sections & enrolment

Publish grades

Unlock published grades

Generate transcripts

View departments

View programs

View & export reports

View analytics

Department Head

Academic quality gatekeeper

The Department Head leads a single department and its instructors. Their signature responsibility is *approving* grades: after an instructor submits marks, the Department Head reviews them and either approves them for publishing or returns them with a written reason for correction. Department Heads can also configure grading for their area, manage the department's courses and sections, and view the analytics that show how their department is doing. They may also step in to enter grades for their department's sections when needed.

Key responsibilities

- Review and approve — or return — submitted grades
- Configure grading components and scale where permitted
- Manage the department's courses and sections
- Monitor the department's academic performance

Representative permissions

View employees

View students

Manage courses

Manage sections & enrolment

Approve/return grades

Configure grading

View departments

View programs

View analytics

View & export reports

Instructor

The person who teaches and grades

The Instructor is at the coalface of the academic year. For every section they are assigned, they record students' marks for each assessment — quizzes, assignments, mid and final exams — watching the system calculate the outcome live as they type. When a section's marks are complete, the Instructor submits them for the Department Head's approval. Instructors work only with their own assigned sections, keeping their view focused and their responsibilities clear.

Key responsibilities

- Enter raw marks for each assessment in assigned sections
- Rely on live, automatic grade calculation while entering
- Submit completed grades for departmental approval

Representative permissions

[View students](#)[Enter grades](#)[Submit grades for approval](#)[Manage courses](#)

Employee

Self-service staff access

The Employee role is the everyday account held by a member of staff who is not part of the administration. It is intentionally modest: an Employee can view their own profile and read their own documents. It gives every staff member a window into the information the college holds about them, without granting access to anyone else's data.

Key responsibilities

- View one's own employee profile
- Make limited edits to one's own details
- Read one's own uploaded documents

Representative permissions

[View own profile](#)[Edit own profile \(limited\)](#)[View/download documents](#)

Student

Self-service learner access

The Student role is the lightest of all. It lets a learner view their own academic information. It is the foundation for a future student portal in which learners will check their own grades and transcripts directly.

Key responsibilities

- View one's own academic information

Representative permissions

View own profile

5.1 Who can do what — the grading relay at a glance

Because grading responsibilities are shared across several roles in a deliberate sequence, the table below shows the chain of authority for the actions that matter most. Each stage can only be performed by the roles marked, which is what makes the grade workflow trustworthy.

Action	Instructor	Dept Head	Registrar	Admin
Enter marks	Yes	Yes*	—	Yes
Submit for approval	Yes	—	—	Yes
Approve / return	—	Yes	—	Yes
Publish (make official)	—	—	Yes	Yes
Unlock a published result	—	—	Yes	Yes
Configure grading scale	—	Yes	—	Yes
Generate transcript	—	—	Yes	Yes

* A Department Head may enter grades for sections within their own department, for example to cover for an absent instructor.

Configurable

This division of duties is enforced by permissions, not hard-coded rules. An administrator can rebalance who holds which permission — or design a brand-new role — entirely through the Administration screens, without any software changes.



III

The Platform, Screen by Screen

A guided tour of every module — from the dashboard to transcripts and analytics.

CHAPTER 6

The Dashboard & Navigation

Every visit to DBPCMS begins at the dashboard and travels through the sidebar. Learn these two elements and you can find anything in seconds.

6.1 Reading your dashboard

The moment you sign in, DBPCMS greets you with a dashboard tailored to your role. This is not a generic welcome screen — it is a live summary of the things that matter to *you*. An administrator sees system-wide totals: how many employees, students, departments and users the college has. A registrar sees academic activity. A department head sees their department's picture. Numbers are presented as clear cards, trends are drawn as friendly charts, and a running activity feed shows the most recent meaningful events across the system. The intent is that within a few seconds of logging in you already know whether anything needs your attention.

6.2 The sidebar — your map of the system

Running down the left of every screen is the sidebar, the master index of everything you can reach. It is **permission-aware**, which is a small phrase with a big consequence: you only see the sections you are allowed to use. An instructor will not see a "System Settings" link, because they cannot manage settings; an HR officer will see "Employees" prominently. This keeps the interface clean and prevents confusion — nobody is tempted by a door they cannot open.

Important

If you were just granted new access but a menu item is still missing, log out and back in. The sidebar is drawn from the permissions stored in your session, which only refresh on a new login.

6.3 Global search

At the top of the screen sits the global search box — the fastest way to jump to any record. Start typing a name, an ID or a code and DBPCMS searches across employees, students, courses and departments simultaneously, returning matching results grouped by type. Like everything else,

search respects your permissions: it will never surface a record you are not allowed to see. For anyone who spends their day looking things up, this is the single most time-saving feature in the platform.

6.4 Language and appearance

The top bar also lets you switch the interface language. DBPCMS ships fully in English and is prepared for Afaan Oromo, chosen to serve the college's community. As the Oromo translations are refined by native speakers, the entire interface can be read in that language at the flick of a switch. A light and dark visual theme are also available, so you can choose whichever is easiest on your eyes.

CHAPTER 7

Academic Structure

Before a single grade can be entered, the college's academic skeleton must exist: departments, the programs they offer, the years and semesters that mark time, and the courses and sections that students actually attend.

7.1 Departments

A department is the largest organising unit in the academic world of DBPCMS. Each department has a name, a unique short code (used across the system as its shorthand), an optional description, and — importantly — a designated head. Departments own almost everything beneath them: the programs students enrol in, the staff who work there, and the courses that are taught. Creating your departments correctly is therefore the very first structural task, because so much else hangs from them.

Name	The full, human-readable department name, e.g. "Information Technology".
Code	A short unique identifier, e.g. "IT", used throughout the system.
Head	The user who leads the department and approves its grades.
Status	Active or inactive — inactive departments are retained for history but hidden from everyday use.

7.2 Programs

A program is the qualification a student pursues — for example a two-year Diploma or a four-year Degree. Every program belongs to exactly one department and records its degree level and its expected duration in years. When you enrol a student, you place them into a program, which is how the system knows what they are studying toward and, ultimately, what their transcript represents.

7.3 Academic years and semesters

Time in DBPCMS is measured in academic years and the semesters within them. An **academic year** has a name and a date range, and exactly one year is marked as the *current* year at any moment — a helpful default that saves you from repeatedly choosing it. Within each year sit **semesters**, each with its own dates, sequence number and a status of *planned*, *active* or *closed*. This structure is what lets the same course be taught again and again across different terms, each offering kept separate and correctly dated.

Tip

Set the current academic year at the start of each new year. Many screens default to it, so keeping it accurate saves everyone a few clicks every single day.

7.4 Courses and sections

A **course** is a subject in the abstract: a code, a title, a number of credit hours and an optional category such as Core or Elective. A **section** is that course made concrete — a specific class, offered in a specific semester, with a section label (such as "A"), an optional capacity, and one or more assigned instructors. Students do not enrol in courses; they enrol in *sections*. This distinction is what allows "Introduction to Programming" to run as three parallel classes in the same term, each with its own instructor and its own list of students.

Assigning instructors

Each section is linked to the instructor or instructors who teach it. This assignment is what later determines whose grade-entry screen the section appears on — an instructor sees only the sections assigned to them, which is both a convenience and a safeguard.

Enrolling students

Enrolment places a student into a section, creating the link to which grades will eventually attach. The system tracks each enrolment's status — enrolled, completed, withdrawn, incomplete or audit — and even records the attempt number, so a student repeating a course is handled cleanly rather than by overwriting their earlier record.

CHAPTER 8

Employee Management

The Employee module is the college's complete, living personnel file — one of the two headline modules of DBPCMS, and the home of the HR office.

8.1 The employee record

Every member of staff is represented by a single, rich employee record. When a new employee is created, DBPCMS automatically assigns them a unique employee number in the form `DBPC-EMP-2026-00001` — the prefix, the year, and a running number that never repeats. The record itself is organised into tabs so that its considerable detail never becomes overwhelming, gathering personal information, employment details, and a set of sub-records into one coherent profile.

Personal	Names, gender, date of birth, nationality, marital status, national ID, tax ID, and contact details.
Employment	Department, position, employment type (full-time, part-time, contract), status, dates of employment and contract end, salary grade, office location and supervisor.
Photo	A profile photograph, uploaded and automatically compressed for efficient storage.

8.2 Sub-records — the depth behind each profile

Beyond the core details, each employee carries four kinds of sub-records, each on its own tab. Together they build a complete professional history.

Education

Every institution attended, the qualification earned, the field of study, the graduation year and the grade achieved — a full academic background for each employee.

Qualifications

Certifications, licences, workshops, training and memberships, each with an issuer, an issue and expiry date, and a reference number. Expiry dates matter: they feed reports that warn when a credential is lapsing.

Employment history

Previous employers, positions held, dates and responsibilities — the career story that preceded the employee's time at the college.

Emergency contacts

The people to reach in an emergency, with their relationship and contact details, kept close at hand for when they are truly needed.

8.3 Documents

Attached to each employee is a secure document store. HR officers can upload scanned degrees, contracts, national IDs, passports, recommendation letters and more. The system validates each file's true type — not merely its name — before accepting it, compresses images to save space, and records its size and a checksum. Files are held through a storage abstraction that means the college can, in future, switch from local storage to a cloud provider without changing how staff use the system at all.

Note

Accepted formats include PDF, Word and Excel documents, and JPEG or PNG images, up to a generous size limit. This covers every kind of official document a college typically files.

8.4 The printable, verifiable profile

With a single click, an authorised user can produce a beautifully formatted, A4 employee profile suitable for official use. Each printed profile carries a QR code linked to a public verification page: anyone — a bank, a government office, another institution — can scan it and instantly confirm that the document genuinely came from Donna Barbar Polytechnic College and has not been forged. This turns an ordinary printout into a trustworthy, tamper-evident credential.

8.5 HR reports

The module includes seven ready-made reports, covered in full in Chapter 13, ranging from a complete staff directory to lists of employees whose contracts are about to expire or who are approaching retirement. Each can be exported to PDF, Excel or CSV.

CHAPTER 9

Student Management & Enrollment

Students are the reason the college exists, and their records sit at the centre of the academic side of DBPCMS.

9.1 The student record

Like employees, every student is one clean record, given an automatic unique identifier in the form `DBPC-STU-2026-00001`. The record captures the student's names, gender, date of birth and contact details, and — crucially — links them to the department and program they belong to. Additional fields record the student's intake batch, their class section label, the semester they are currently in, and their status: active, graduated, withdrawn or suspended. A profile photograph can also be uploaded, which later appears on their transcript.

Identity	Auto-generated student number, full name, gender, date of birth, contact details.
Academic placement	Department, program, batch, class section and current semester.
Status	Active, graduated, withdrawn or suspended — tracking the student's journey.
Photo	A portrait used on the official transcript.

9.2 Enrollment into sections

Placing a student into their classes is done through **enrollment**. From a section, the registrar adds the students who will take that class, and from a student's record you can see every section they are enrolled in. Each enrollment is the precise point to which grades attach later — so accurate enrolment is the foundation of accurate grading. The system records each enrolment's status and attempt number, handling repeats and withdrawals gracefully without ever losing the earlier attempt.

Tip

Because a grade is attached to an *enrollment* rather than directly to a student, a learner who repeats a course keeps both attempts on record — the system simply increments the attempt number. This is exactly what a fair, auditable academic record requires.

9.3 Student photos

Uploading a student's photograph does more than personalise their record: the same image is placed on their official transcript, making the document harder to misuse and easier to match to its rightful owner. As with employee photos, images are automatically compressed on upload so that storage stays efficient even across thousands of students.

CHAPTER 10

Grading — Setup & Configuration

Before any marks are entered, the grading engine must know the college's rules: how assessments are weighted, and how percentages become letter grades. Both are set entirely in the interface.

10.1 A grading engine that bends to the college

One of the founding promises of DBPCMS is that the way grades are calculated is never buried in the software. Instead, it lives on configuration screens that an authorised user can adjust. If the college decides next year that the final exam should be worth 60% instead of 50%, or that the pass mark should change, that is a few clicks — not a software project. This chapter describes the values the system ships with, and how to change them.

10.2 Grade components and their weights

A **grade component** is one type of assessment. DBPCMS ships with four, whose weights add up to 100%. Instructors enter a raw mark for each component — for example a score out of 10 for a quiz — and the system converts and combines them automatically. The default components are:

Component	Weight	Maximum raw score	Meaning
Quiz	10%	10	Short in-class or take-home assessments.
Assignment	15%	15	Homework, projects and continuous coursework.
Mid Exam	25%	25	The mid-semester examination.
Final Exam	50%	50	The comprehensive end-of-semester examination.

Configurable

The name, weight and maximum score of every component are editable, and you may add or remove components entirely. The only rule the system enforces is that the weights of all active components must total 100%.

10.3 The grading scale

The **grading scale** is the ladder that turns a final percentage into a letter grade and a grade point on the 4.0 scale. Each rung — called a *band* — covers a percentage range and declares its letter, its grade point, and whether it counts as a pass. The default college-wide scale is shown below.

Letter	Percentage range	Grade point	Outcome
A+	90 – 100	4.00	Pass
A	85 – 89	4.00	Pass
A-	80 – 84	3.75	Pass
B+	75 – 79	3.50	Pass
B	70 – 74	3.00	Pass
B-	65 – 69	2.75	Pass
C+	60 – 64	2.50	Pass
C	50 – 59	2.00	Pass
D	40 – 49	1.00	Fail
F	0 – 39	0.00	Fail

10.4 Pass mark, rounding and GPA

Two more settings complete the picture. The **pass mark** — 50% by default — is the percentage at or above which a student passes a course. The **rounding rule** decides how a borderline percentage is treated; the default, "half up", rounds a value such as 64.5 up to 65. From the resulting grade points and each course's credit hours, DBPCMS computes the Grade Point Average on the familiar 4.0 scale, both for a single semester and cumulatively across a student's whole time at the college.

Configurable

The pass mark, the rounding rule, and every band's range, letter and grade point can all be edited on the grading setup screen. For Version 1 a single college-wide scale is used; the system is already built to support per-department scales in future.

CHAPTER 11

Grading — Entry, Approval & Publishing

This is where grades come to life — and where DBPCMS's most carefully designed workflow ensures that every published result is accurate, reviewed and permanent.

11.1 Entering marks

An instructor opens the grade-entry screen and sees only the sections assigned to them. Choosing a section reveals a grid: one row per enrolled student, one column per grade component. The instructor types each student's *raw* mark — a score out of that component's maximum, such as 8 out of 10 for a quiz — and, as they type, the system instantly computes the weighted total, the percentage, the letter grade and the pass/fail outcome. There is no separate "calculate" step; the result is always live and visible. To protect against lost work, entries are saved automatically a moment after you stop typing.

Tip

Enter the actual mark out of the component's maximum, not out of 100. The system knows each component's maximum and does the conversion for you — so a quiz mark is simply "8", never "80".

11.2 The approval workflow — a relay of trust

A grade in DBPCMS is not official the moment it is typed. It travels through a deliberate sequence, with each stage performed by a different role, so that no single person can quietly determine a student's result. This is the heart of the system's academic integrity.

1**Enter & Submit — Instructor**

The instructor completes all marks for a section and submits them. The section's status moves from *draft* to *submitted*, and the marks lock against further casual editing.

2**Approve or Return — Department Head**

The department head reviews the submitted marks. They either *approve* them, advancing the section, or *return* them with a written reason, sending them back to the instructor for correction.

3

Publish — Registrar

The registrar performs the final, decisive act: publishing. At this moment the results become official and a permanent, immutable snapshot is frozen.

4

Locked

Published results are locked. They now appear on transcripts and can no longer be changed through ordinary editing — only through a deliberate, authorised unlock.

11.3 Returning grades for correction

When a department head spots a problem, they do not silently change the marks — they return the section with a reason. The instructor sees exactly why it came back, makes the correction, and re-submits. This keeps responsibility where it belongs and leaves a clear trail of what happened and why.

11.4 The immutable snapshot

Publishing does something subtle but powerful: it freezes a snapshot of the grading rules that were in force at that moment — the components, weights and scale — and stores it alongside the result. This means that even if the college later changes its grading scale, a previously published result remains exactly as it was calculated, forever. A transcript printed today and a transcript printed in ten years will always agree.

11.5 Unlocking for genuine corrections

Occasionally a legitimate error surfaces after publishing. For these cases, and these cases only, the registrar or an administrator can **unlock** a published result, allowing a correction and re-publishing. Every unlock is recorded in the audit log, so this powerful ability is always accountable.

Important

Unlocking a published grade is an exceptional action, not a routine one. Because published results feed transcripts and official records, an unlock should always be tied to a documented reason.

CHAPTER 12

Transcripts

The transcript is the college's most important academic document — the formal, verifiable summary of everything a student has achieved. DBPCMS produces it automatically from published grades.

12.1 What a transcript contains

DBPCMS transcript brings together, on one official page, the student's identity and photo-
A graph, the courses they have taken grouped by semester, the grade earned in each, and the resulting Grade Point Averages. It shows both the **semester GPA** — how the student performed in a single term — and the **cumulative GPA**, the running average across their entire studies. Because it is generated from published, frozen results, it is always accurate and internally consistent.

12.2 Generating and browsing

The registrar can reach any student's transcript in two ways: by searching directly for the student, or by browsing down through department, program and section to find them. This dual approach suits both the moment when you know exactly who you want and the moment when you are working through a whole class.

12.3 Printing and verification

Transcripts print to a clean, formal A4 layout ready for official issue. Like the employee profile, each carries a QR code that links to a public verification page. A university, employer or embassy receiving the document can scan the code and confirm, independently, that it was genuinely issued by the college — a powerful defence against forgery that turns a piece of paper into a checkable credential.

Note

Only *published* results appear on a transcript. Grades still moving through the entry, approval or return stages are correctly excluded until they become official — so a transcript never shows a provisional mark.

CHAPTER 13

Reports & Analytics

Data is only valuable when it answers questions. DBPCMS turns its records into ready-made reports and analytics, each exportable to the format you need.

13.1 HR reports

The Employee module offers seven built-in reports that answer the questions the HR office and leadership ask most often. Each can be filtered as appropriate and exported as a PDF for printing, an Excel workbook for further analysis, or a CSV file for use in other tools.

Report	What it tells you
Employee Directory	A complete listing of all staff — the master roll of the college.
Department Employees	Staff grouped by the department they work in, revealing how people are distributed.
Qualification Summary	An overview of the qualifications held across the workforce.
New Employees	Recently hired staff, useful for onboarding and induction.
Contract Expiry	Employees whose contracts are approaching their end date, within a configurable window — so renewals are never missed.
Retirement List	Employees approaching the configurable retirement age, for succession planning.
Staff Distribution	A breakdown of how staff are spread across departments and types.

Configurable

The retirement age and the contract-expiry warning window are both system settings. Change them once and the Retirement List and Contract Expiry reports adjust automatically.

13.2 Academic analytics

On the academic side, DBPCMS analyses published results to reveal how teaching and learning are going. These insights are aimed at deans, registrars and department heads who need to see patterns rather than individual marks.

Pass rate	The proportion of students passing, by course, section or department.
Top students	The highest achievers, useful for recognition and awards.
Department GPA	A comparison of average performance across departments.
Failure analysis	Where students are struggling most, so support can be directed effectively.

13.3 Exporting

Every report can leave the system in one of three formats. Choose **PDF** when you want a clean, printable document to file or share; **Excel** when you want to sort, filter or chart the data yourself; and **CSV** when you need to feed the figures into another program. The exporters are shared across the whole platform, so every report you produce carries a consistent, professional look.



IV

Administration & Operations

Configuring, safeguarding and understanding the system over the long term.

CHAPTER 14

System Settings & Configuration

The Settings screen is where the college's rules are written. It is the practical expression of the platform's guiding promise: everything that could reasonably change is changeable here, not hidden in code.

14.1 Why nothing is hard-coded

A college's rules evolve. Retirement ages are revised, ID formats are tidied, the institution's own name might be adjusted for a formal document. In many systems each such change means calling a programmer. DBPCMS was deliberately built to avoid that. The values that govern its behaviour are stored as editable settings, and the business logic reads those settings rather than fixed numbers. The result is a system the college genuinely owns and can adjust itself.

14.2 The editable settings

Setting	Default	What it controls
Institution name	Donna Barbar Polytechnic College	The name printed on official documents and reports.
Retirement age	60	The age at which staff appear on the Retirement List report.
Contract expiry window	60 days	How far ahead the Contract Expiry report looks for ending contracts.
Student ID prefix	DBPC-STU	The prefix for auto-generated student numbers, e.g. DBPC-STU-2026-00001.
Employee ID prefix	DBPC-EMP	The prefix for auto-generated employee numbers, e.g. DBPC-EMP-2026-00001.

Alongside these, the grading configuration described in Chapter 10 — components, weights, the grading scale, pass mark and rounding — is itself a form of configuration, editable by those with

the grading permission. Between them, these screens let the college shape the system to its own practices without ever touching the software.

Important

Settings changes take effect immediately and affect everyone. Changing an ID prefix, for instance, alters the format of all *future* generated IDs, while existing records keep the IDs they were given. Treat this screen with the same care as any other powerful tool.

CHAPTER 15

The Audit Log

If the audit log had a motto, it would be "the system never forgets". It is the complete, honest history of who did what.

15.1 What the log records

Whenever a meaningful change is made — a record created, updated or archived, a login, a password reset, a grade published — DBPCMS quietly writes an entry to the audit log. Each entry captures the responsible user, the exact time, the type of action, the record affected, and, for changes, the values *before* and *after*. This happens automatically and cannot be switched off by ordinary users, which is precisely what makes it trustworthy.

15.2 Investigating changes

Administrators can open the audit-log viewer to trace the history of any record or the activity of any user. If a question ever arises — "when did this employee's status change, and who changed it?" or "who published these grades?" — the answer is there, in black and white. This transforms the system from a mere store of current facts into an accountable record of how those facts came to be, which is invaluable for governance, dispute resolution and simple peace of mind.

Note

Because the audit log combines with soft deletes — where records are archived rather than destroyed — DBPCMS can reconstruct not just *what* the data is today, but the full story of how it got there.

CHAPTER 16

Backups & Data Protection

The most valuable thing in DBPCMS is not the software — it is the data. This chapter explains how that data is protected and how it can be recovered.

16.1 The backup routine

DBPCMS includes a tested backup routine that captures the entire database — every employee, student, grade, document reference and setting — into a single dated file. Run regularly, these backups mean that even in the worst case, a hardware failure or a serious mistake, the college can return to a recent, complete copy of its records. A backup is a form of insurance: unremarkable until the day it saves everything.

Tip

Establish a rhythm — for example a backup at the end of each working day, and an extra one before any major change such as a software update or a bulk data import. The more recent your latest good backup, the less you can ever lose.

16.2 Restoring from a backup

Restoring reverses the process: it takes a backup file and rebuilds the database from it, returning the system to exactly the state it was in when that backup was taken. This is a deliberate, administrator-only operation, because it replaces the current data entirely. Used correctly — after a serious problem, or to recover a mistakenly lost record — it is the final, reassuring safety net beneath everything else the system does.

Important

A restore overwrites the current database with the contents of the backup. Always be certain you are restoring the right file, and consider taking a fresh backup of the current state first, so that no path is ever closed to you.

16.3 Storage of documents

Uploaded documents and photographs are held through a storage layer designed to be swapped without disruption. Today they live on the server's local storage; in future the college can move to a cloud provider for extra resilience, and the change will be invisible to everyone who uses the system. When you plan backups, remember to include this document store alongside the database so that files and their records are always protected together.

CHAPTER 17

Frequently Asked Questions

The everyday questions that come up most often, answered plainly.

I just got new access, but I still can't see the new menu. Why?

Your permissions are carried in your login session. Log out and log back in, and the new menu items will appear. This single step solves the great majority of "I can't see it" questions.

I forgot my password. What do I do?

Contact a System Administrator. DBPCMS does not send reset emails; instead, an administrator sets a temporary password for you, which you will be required to change the next time you log in.

Why can't I publish grades? I entered them myself.

Entering and publishing are deliberately different responsibilities. Instructors enter and submit; department heads approve; only the registrar (or an administrator) publishes. This separation is what makes every published result trustworthy — see Chapter 11.

A student needs a grade corrected after it was published. Is that possible?

Yes, but carefully. A registrar or administrator can unlock the published result, allow the correction, and re-publish. Every unlock is recorded in the audit log, so the correction is always accountable.

Do I enter quiz marks out of 100?

No — enter the raw mark out of that component's maximum. If a quiz is out of 10 and a student scored 8, you type "8". The system handles the weighting and conversion for you.

I deleted a record by mistake. Is it gone forever?

Almost certainly not. Most deletions are "soft" — the record is archived and hidden rather than destroyed. Ask an administrator, who can recover it, and consult the audit log for the details.

Can the college change the grading scale or pass mark?

Absolutely. Both live on the grading configuration screen and can be edited by an authorised user. Results already published keep the rules that were in force when they were published, thanks to the immutable snapshot described in Chapter 11.

How do I check that a printed transcript or profile is genuine?

Scan the QR code on the document. It leads to a public verification page that confirms whether the document was genuinely issued by the college — no login required.

What languages does the system support?

English is complete, and Afaan Oromo support is built in and being refined. You can switch language from the top bar at any time.

CHAPTER 18

Glossary of Terms

Every important term used in this manual, defined in one place for quick reference.

Academic Year	A named year of study with start and end dates; one is marked current at a time.
Audit Log	The automatic, permanent record of who did what and when, including before-and-after values.
Band	One rung of the grading scale, covering a percentage range and defining its letter and grade point.
Course	A subject in the abstract — code, title and credit hours.
Cumulative GPA	The running grade point average across a student's entire studies.
Department	The largest academic unit; owns programs, staff and courses.
Enrollment	The link placing one student in one section; the thing a grade attaches to.
Employee	A member of staff with a full personal and professional profile.
Grade Component	One assessment type — Quiz, Assignment, Mid Exam, Final Exam — with a weight and maximum.
Grading Scale	The set of bands turning a percentage into a letter grade and grade point.
Grade Point Average (GPA)	A weighted average of grade points on the 4.0 scale.
Immutable Snapshot	The frozen copy of grading rules stored with a result when it is published.
Permission	The right to perform one specific action, e.g. publish grades.
Program	A qualification a student studies toward, belonging to a department.
Record	One individual example of an entity, e.g. a single student.
Role	A named bundle of permissions given to people who do a similar job.
Section	A specific offering of a course in a specific semester — the actual class.
Semester	A teaching period within an academic year, with its own dates and status.

Soft Delete	Archiving a record — hiding rather than destroying it — so history is preserved.
Student	A learner enrolled at the college, belonging to a department and program.
Transcript	The official, verifiable summary of a student's courses and grades.
User	A login account, granted one or more roles, used to access the system.
Version (optimistic lock)	A hidden number preventing two people from silently overwriting each other's edits.

— End of Manual —

Donna Barbar Polytechnic College Management System · Version 1.0 · 2026

This handbook describes the system as delivered. Because DBPCMS is designed to be configured by the college, some values shown as defaults may differ on your installation.